Frequency-Domain and Volatility Regime Filters for Time-Series Momentum: A Cell-Conditional Decomposition

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Abstract

I tested whether a frequency-domain regime indicator, derived from the rolling power spectral density of returns, adds value beyond a realized-volatility regime indicator when used to filter a time-series momentum strategy on a 10-asset ETF universe (2010–2026). The two regime classifiers are empirically independent (mask agreement $\approx 50\%$), yet neither filter alone outperforms a vol-filtered baseline. Their intersection does. A cell-conditional decomposition shows that momentum's risk-adjusted edge is concentrated in the cell where both filters classify the asset as favorable (cell Sharpe 0.60 versus next-best 0.15). The intersection-filtered strategy is the only strategy in the comparison set whose Sharpe is stable across in-sample, out-of-sample, and sub-period evaluations (range 0.51 to 0.66). Memmel (2003) tests for Sharpe differences are non-significant, and the result is sensitive to PSD window length.

1 Introduction

Time-series momentum (TSMOM) is among the best-documented systematic trading strategies in the literature, with positive risk-adjusted returns established across asset classes and decades [4, 6]. Momentum's performance is widely understood to be regime-dependent, and volatility-based filtering of momentum positions is a well-established refinement [8]. Whether volatility-based filtering exhausts the useful information available from each asset's return history is an empirical question that has received less attention.

This project investigates a frequency-domain alternative. The power spectral density of returns describes how variance is distributed across timescales and is therefore sensitive to the persistence structure of returns, which is conceptually distinct from the level of volatility. I constructed a regime indicator from the rolling spectral density, applied it as a filter to a standard TSMOM strategy, and compared against both unfiltered and volatility-filtered baselines on a 10-asset ETF universe. The two regime classifiers turned out to be empirically independent, neither single filter dominates the other, but their intersection produced the most stable risk-adjusted performance across in-sample, out-of-sample, and sub-period evaluations. A cell-conditional decomposition explains the result: momentum's edge is concentrated in the joint cell where both filters classify the asset as favorable. Section 2 describes the strategies; Section 3 the methodology; Section 4 the empirical findings; Section 5 the robustness checks; and Section 6–7 discussion and limitations.

2 Strategy Construction

This section introduces the four trading strategies evaluated in the project. Each strategy is built from the same underlying time-series momentum signal and the same vol-scaled position sizing; they differ only in the regime filter applied. The construction details for all four strategies are described in Section 3; this section focuses on the rationale for each layer.

2.1 Layer 0: Unconditional Time-Series Momentum

The baseline strategy is unconditional time-series momentum (TSMOM) following [4]. For each asset and at each month-end, the strategy goes long if the trailing 12-month minus 1-month return is positive, short if it is negative, and flat otherwise. Position sizes are vol-scaled to a 10% annualized target so each asset contributes roughly equal risk to the portfolio. The strategy trades every month regardless of market conditions and applies no regime filter.

Unconditional TSMOM is well-established academically and has been documented to deliver positive risk-adjusted returns across decades and across asset classes [6]. It serves here as the control against which the regime-filtered variants are measured. If a conditioning strategy fails to improve on this baseline, then it would not be considered useful.

2.2 Layer 1: PSD-Filtered TSMOM

The first regime-filtered variant I applied uses the frequency-domain regime mask m^{psd} defined in Section 3.3. The strategy takes the unconditional TSMOM position only when the asset is currently classified as in a **trending regime** ($z^{\text{psd}} > 0$); in choppy regimes, the position is set to zero.

The motivation for this strategy is the project's central hypothesis: because momentum strategies profit from persistent directional movement, and because the PSD log-ratio is theoretically sensitive to the autocorrelation structure that produces such persistence, conditioning momentum trading on the PSD-based regime classification should improve risk-adjusted returns. The PSD-filtered strategy is the direct empirical test of this hypothesis.

2.3 Layer 2: Vol-Filtered TSMOM

The second regime-filtered variant applies the volatility-based regime mask m^{vol} defined in Section 3.3, taking the unconditional TSMOM position only when the asset is in a low-volatility regime ($z^{\text{vol}} \leq 0$) and going flat otherwise.

This strategy is included for a specific methodological purpose: it isolates what the frequency-domain information contributes. Volatility-based filtering of momentum strategies is well-documented in the literature, most notably in [8], and represents the standard practitioner approach to regime conditioning. If the PSD-filtered strategy fails to improve on the vol-filtered baseline, then the frequency-domain information adds nothing beyond what realized volatility already provides. The vol-filtered strategy is therefore both a competing approach and a methodological benchmark that the PSD filter must clear to demonstrate distinct value.

The two regime classifiers are constructed in parallel (see Section 3.3) so that any performance difference reflects the differences in the content of the signals rather than artifacts of construction.

2.4 Layer 3: Intersection-Filtered TSMOM

The final strategy applies both filters jointly: the unconditional TSMOM position is taken only when the asset is simultaneously classified as trending by the PSD filter and as low-volatility by the vol filter. In all other configurations, the position is zero. Formally, the intersection mask is $m_{i,t}^{\text{int}} = m_{i,t}^{\text{psd}} \cdot m_{i,t}^{\text{vol}}$, applied multiplicatively to the unconditional position as in Equation 10.

Two empirical findings motivate this strategy, both presented in Section 4. First, the PSD and volatility regimes are empirically independent on this universe: the two masks agree on approximately 49% of $(\text{date}, \text{asset})$ cells, and the per-asset correlations of the mask time series are within ± 0.10 of zero. The two classifiers measure substantively distinct aspects of market state. Second, when the unconditional momentum strategy is restricted to each of the four cells of the 2×2 regime partition, risk-adjusted performance is concentrated in the cell where both filters classify the asset as favorable (trending $\cap$ low-volatility). The intersection-filtered strategy isolates this cell.

The intersection strategy is conservative, i.e., it trades in approximately 25% of $(\text{date}, \text{asset})$ cells rather than the 50% of either single-filter strategy. The economic question it answers is whether the higher concentration on favorable regimes more than compensates for the lower in-market exposure.

2.5 Summary

The four strategies are summarized in Table 1. All share the same underlying TSMOM signal and vol-scaled position sizing; they differ only in the regime filter, which determines when the position is taken versus held flat.

Strategy	**Regime filter**	**In-market when**
Layer 0: Unconditional	none	always
Layer 1: PSD-filtered	m^{psd}	$z^{\text{psd}} > 0$
Layer 2: Vol-filtered	m^{vol}	$z^{\text{vol}} \leq 0$
Layer 3: Intersection	$m^{\text{psd}} \cdot m^{\text{vol}}$	$z^{\text{psd}} > 0$ and $z^{\text{vol}} \leq 0$

Table 1: The four strategies evaluated. All four use the same underlying momentum signal $s_{i,t}$ (Equation 8) and the same vol-scaled positioning (Equation 9); they differ only in the multiplicative regime mask applied via Equation 10.

3 Methodology

The data, the spectral feature that powers my regime classifier, the construction of the regime masks themselves, the backtest protocol, and the statistical tests used to compare strategies are described here. The four strategies evaluated are defined in Section 2; here, I only describe the methodological infrastructure common to all of them.

3.1 Data and Universe

I used the daily adjusted closing prices for ten exchange-traded funds (ETFs) spanning the major liquid asset classes: U.S. large-cap equities (SPY), U.S. large-cap technology (QQQ), U.S. small-cap equities (IWM), developed international equities (EFA), emerging-market equities (EEM), long-duration U.S. Treasuries (TLT), intermediate U.S. Treasuries (IEF), gold (GLD), a broad commodity basket (DBC), and U.S. real estate (VNQ). Prices are obtained from Yahoo Finance and span January 2007 through the most recent available date. I used the 2007 start date for approximately three years of warm-up before the formal backtest begins in January 2010. This provided sufficient data to populate the rolling windows of all indicators.

I used ETFs rather than futures for two reasons: data availability is uniform and free, and the resulting strategy is directly implementable by retail or institutional investors without futures account infrastructure. The principal cost of this choice is that ETF-based time-series momentum strategies have historically delivered lower Sharpe ratios than equivalent futures-based strategies [6]. I have discussed this limitation explicitly in Section 7.

Returns are computed as daily log returns,

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right), \tag{1}$$

where $P_{i,t}$ is the adjusted close of asset i on day t. I used log returns instead of arithmetic returns as they are additive over time, and this simplified the spectral analysis described below. I also dropped any trading day on which any asset has missing data.

3.2 Spectral Feature: Rolling PSD Log-Ratio

The central indicator of my regime classifier is the **rolling power spectral density (PSD) log-ratio** of returns. Intuitively, the PSD describes how the variance of a stochastic process is distributed across frequencies (or timescales). A process whose returns exhibit positive short-lag autocorrelation concentrates spectral power at low frequencies, while a mean-reverting process concentrates power at high frequencies. The PSD log-ratio quantifies this concentration.

Welch's method. For a window of $N = 60$ daily returns ending on day t for asset i, I estimated the PSD using Welch's method [3]. Welch's estimator divides the window into overlapping segments, computes a modified periodogram on each, and averages the periodograms. I chose this over the raw periodogram as Welch's estimator trades frequency resolution for variance reduction, which is appropriate here because power over broad frequency bands is integrated, rather than resolving narrow spectral peaks. I also set the segment length equal to the window (60 days), 50% segment overlap, linear detrending applied to each segment and density-normalized output.

The estimator returns power at frequencies $f_k = k/N$ cycles per day, for $k = 1, \ldots, N/2$. I excluded the zero-frequency component, which after detrending is by construction near zero.

Frequency partition. I partitioned the frequency axis at a cutoff period of $\tau_c = 20$ days, corresponding to cutoff frequency $f_c = 1/\tau_c = 0.05$ cycles per day. The low-frequency band contains frequencies $f < f_c$ (periods longer than 20 days); the high-frequency band contains $f \geq f_c$ (periods shorter than 20 days, up to the Nyquist limit of 0.5 cycles per day).

The choice of $\tau_c = 20$ days is motivated by the timescale of my underlying trading strategy. My momentum signal evaluates trailing 12-month versus 1-month returns and rebalances monthly. Cycles with period shorter than approximately one month are noise relative to this signal; cycles longer than one month are the persistence the signal is designed to exploit. Thus, the cutoff separates the frequency content the strategy seeks to capture from the frequency content it should ignore. This is an economic justification, not a statistical optimization; I treated τ_c as a fixed hyperparameter and report sensitivity to this choice in Section 5.

Log-ratio feature. Let $S_i(f \mid t)$ denote the Welch PSD estimate for asset i on day t. Define the integrated low-band and high-band powers as

$$L_{i,t} = \sum_{f_k < f_c} S_i(f_k \mid t), \qquad H_{i,t} = \sum_{f_k \geq f_c} S_i(f_k \mid t). \tag{2}$$

The PSD log-ratio feature is

$$\phi_{i,t} = \ln\left(\frac{L_{i,t}}{H_{i,t}}\right). \tag{3}$$

I used the log transformation rather than the raw ratio because:

i. The raw ratio is bounded below by zero but unbounded above, producing strongly right-skewed values that destabilize subsequent z-score normalization.

ii. The log ratio is symmetric around zero in the sense that a doubling of low-band power and a halving of low-band power produce equal-magnitude opposite-sign changes in ϕ.

Validation on synthetic data. Before applying the selected PSD-based feature to market data, I verified its behavior on synthetic series with known autocorrelation structure. White noise produces ϕ stable around a baseline value reflecting the asymmetric width of the two frequency bands. An AR(1) process with positive coefficient produces systematically higher ϕ; an AR(1) with negative coefficient produces systematically lower ϕ. The ordering matches the theoretical prediction from the Wiener–Khinchin relation between autocorrelation and spectral density.

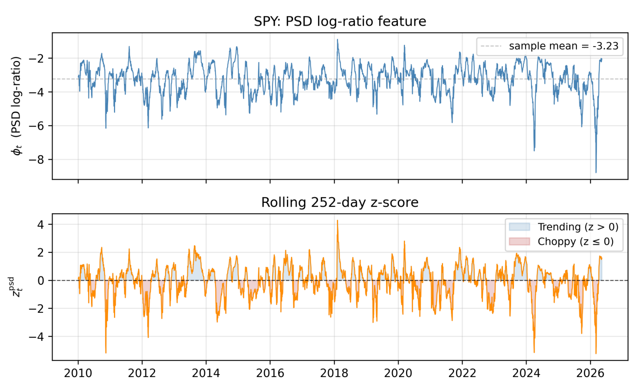

Figure 1: Rolling 60-day PSD log-ratio feature ϕ_t for SPY (top), and its rolling 252-day z-score z_t (bottom). The z-score is the classifier input: $z_t > 0$ corresponds to the trending regime.

3.3 Regime Classification

The PSD feature $\phi_{i,t}$ is continuous; I converted it to a binary regime label via rolling z-score normalization. For each asset, define

$$z_{i,t}^{\text{psd}} = \frac{\phi_{i,t} - \mu_{i,t}^{\text{psd}}}{\sigma_{i,t}^{\text{psd}}}, \tag{4}$$

where $\mu_{i,t}^{\text{psd}}$ and $\sigma_{i,t}^{\text{psd}}$ are the sample mean and standard deviation of $\phi_{i,\cdot}$ over the trailing 252 trading days ending on day t. The PSD regime label is

$$m_{i,t}^{\text{psd}} = \begin{cases} 1 & \text{if } z_{i,t}^{\text{psd}} > 0 \quad \text{(trending)} \\ 0 & \text{otherwise} \quad \text{(choppy).} \end{cases} \tag{5}$$

I chose the 252-day normalization window to approximate one trading year, long enough to capture multiple regime transitions but short enough that the classifier remains responsive. Both the feature and its z-score use only data available strictly on or before day t, so the resulting classification carries no look-ahead bias.

Volatility regime baseline. To isolate the contribution of the frequency-domain information, I constructed a parallel regime classifier based on realized volatility. Let

$$\nu_{i,t} = \sigma(r_{i,t-19}, \ldots, r_{i,t}) \cdot \sqrt{252} \tag{6}$$

denote annualized 20-day realized volatility, and let $z_{i,t}^{\text{vol}}$ be its trailing-252-day z-score, constructed identically to Equation 4. The vol regime label is

$$m_{i,t}^{\text{vol}} = \begin{cases} 1 & \text{if } z_{i,t}^{\text{vol}} \leq 0 \quad \text{(low-vol)} \\ 0 & \text{otherwise} \quad \text{(high-vol).} \end{cases} \tag{7}$$

The inverted sign convention relative to the PSD mask I have used is: the favorable regime for momentum is the *low*-vol regime, so the mask is 1 when z^{vol} is below its trailing mean.

The two regime classifiers are constructed in parallel. Both operate per asset, both use 252-day self-normalization, both produce binary classifications, and both use exclusively trailing data. I believe that any empirical difference in their performance would reflect the difference in what is being measured (frequency-domain structure versus realized volatility level), not artifacts of construction.

3.4 Pre-Registration of Parameters

All hyperparameters governing the regime classifier and the underlying trading strategy were fixed in a written pre-registration document before any backtesting was performed. The pre-registered values are:

- PSD window length (60 days)
- Welch segment overlap (50%)
- Frequency cutoff period (20 days)
- Z-score normalization window (252 days)
- Regime classification threshold ($z = 0$)
- Momentum signal lookback (12 months minus 1 month)
- Position vol target (10% annualized)
- Volatility estimation window for position sizing (60 days)
- Rebalance frequency (monthly)
- Transaction cost assumption (5 basis points per unit of turnover)
- Backtest start date (January 1, 2010)

Each value is justified on economic or statistical grounds independent of in-sample performance. The pre-registration document, reproduced in Section A, is dated prior to results being computed; this discipline protects against the well-documented tendency for in-sample parameter optimization to produce inflated performance estimates that fail to generalize [2]. I treated the pre-registered values as fixed throughout the main analysis and report sensitivity to perturbations in Section 5.

3.5 Backtest Protocol

Position construction. For each asset and each rebalancing date, the unconditional time-series momentum signal is

$$s_{i,t} = \operatorname{sign}\left(r_{i,t}^{(12m)} - r_{i,t}^{(1m)}\right), \tag{8}$$

where $r_{i,t}^{(k)}$ denotes the trailing k-month log return.

Signals take values in $s_{i,t} \in \{-1, 0, +1\}$: the strategy goes long the asset when its trailing 12-month minus 1-month return is positive, short when it is negative, and flat at exactly zero (a measure-zero case in practice). The 12-month minus 1-month construction follows [4] and removes the short-horizon reversal effect documented in [5].

Short positions are assumed implementable at the prevailing closing price, a reasonable approximation for the highly liquid ETFs in my universe but one that ignores borrow costs, locate fees and any operational frictions specific to shorting. I have discussed this in Section 7.

Raw signals are converted to vol-scaled positions:

$$w_{i,t} = s_{i,t} \cdot \frac{\sigma^*}{\hat{\sigma}_{i,t}}, \tag{9}$$

where $\sigma^* = 0.10$ is the per-asset vol target and $\hat{\sigma}_{i,t}$ is annualized 60-day realized vol. Vol scaling ensures each asset contributes approximately equal risk to the portfolio, preventing high-vol assets from dominating.

Regime conditioning. A regime-filtered strategy applies a multiplicative mask to the position,

$$w_{i,t}^{\text{filtered}} = w_{i,t} \cdot m_{i,t}, \tag{10}$$

where $m_{i,t}$ is one of m^{psd}, m^{vol}, or $m^{\text{psd}} \cdot m^{\text{vol}}$ (the intersection mask). The unconditional strategy uses $m_{i,t} \equiv 1$.

Rebalancing and look-ahead protection. Positions are computed at the close of the last trading day of each calendar month and held constant through the next month. To eliminate any possibility of look-ahead bias, monthly positions are explicitly lagged by one trading day, so the position held on day t uses information available strictly before t.

Portfolio return and costs. Daily portfolio return is the equal-weighted average of asset-level returns under the chosen positions:

$$R_t = \frac{1}{N_{\text{assets}}} \sum_{i=1}^{N_{\text{assets}}} w_{i,t-1} \cdot \left(e^{r_{i,t}} - 1\right) - c_t, \tag{11}$$

where the equal weighting reflects the equal risk-target normalization applied at the asset level, and

$$c_t = \kappa \cdot \frac{1}{N_{\text{assets}}} \sum_i |w_{i,t-1} - w_{i,t-2}| \tag{12}$$

is the transaction cost charge with $\kappa = 5$ basis points. The cost parameter is conservative for the highly liquid ETFs in my universe.

3.6 Performance Metrics

For each strategy and each evaluation period, I reported the following performance metrics:

- annualized return daily mean $\times 252$)
- annualized volatility (daily standard deviation $\times \sqrt{252}$)
- Sharpe ratio (annualized return divided by annualized volatility, with zero risk-free rate)
- maximum drawdown (largest peak-to-trough decline in cumulative log return)
- Calmar ratio (annualized return divided by the absolute value of maximum drawdown)
- monthly hit rate (fraction of calendar months with positive total return)

The zero risk-free rate convention is standard when comparing strategies that share the same risk-free benchmark; any common subtraction would shift all reported Sharpes by the same amount and would not affect rankings.

3.7 Statistical Tests

Sharpe ratio standard errors. I used the standard approximation from [1],

$$\mathrm{SE}(\widehat{\mathrm{SR}}) \approx \sqrt{\frac{1 + \frac{1}{2}\widehat{\mathrm{SR}}^2}{T_y}}, \tag{13}$$

where T_y is the effective sample length in years. This expression clarifies that individual Sharpe estimates from subsamples carry substantial sampling error even on multi-year data.

Sharpe difference tests. To compare two strategies evaluated on contemporaneous returns, I applied the Memmel (2003) test for the difference of Sharpe ratios [7]. For two return series r_1, r_2 with n joint observations, sample Sharpes $\widehat{SR}_1, \widehat{SR}_2$ and sample correlation ρ, the test statistic is

$$z = \frac{\widehat{SR}_1 - \widehat{SR}_2}{\sqrt{\frac{1}{n}\left(2 - 2\rho + \frac{1}{2}(\widehat{SR}_1^2 + \widehat{SR}_2^2 - 2\widehat{SR}_1\widehat{SR}_2\rho^2)\right)}}, \tag{14}$$

Power considerations. The Memmel test, like all Sharpe comparison tests, has limited power on the sample sizes typical of finance backtests. I reported p-values throughout but interpret them carefully: a non-significant difference between two Sharpes **does not** imply they are equal, only that the data cannot distinguish them. This limitation is intrinsic to the estimation problem rather than to my strategy.

4 Results

This section presents the empirical findings of the project. I began with the regime independence diagnostic that motivated the rest of the analysis, then turned to the headline strategy comparison, the cell-conditional decomposition that explains the strategy rankings, and the statistical tests that show my conclusions. Robustness across sub-periods and the out-of-sample evaluation are done in Section 5.

4.1 Regime Independence

The two regime classifiers, m^{psd} and m^{vol}, are constructed parallelly (Section 3.3) so that any difference in their behavior reflects the differences in what they measure. The first empirical question is whether they measure the same underlying phenomenon by different routes, or whether they capture distinct aspects of market state.

Table 2 reports the fraction of (date, asset) cells on which the two masks agree, alongside the per-asset correlation of the two mask time series. On the full sample, the two masks agree on 50.6% of cells, essentially indistinguishable from the 50% baseline that independent random masks with equal marginals would produce. Per-asset correlations range from -0.068 (SPY) to +0.105 (IEF), all consistent with zero.

	Mask agreement	**PSD in-market**	**Vol in-market**
Full sample (2010–2026)	50.6%	52.3%	61.6%

Table 2

The slight skew in the vol mask (61.6% in-market versus the symmetric expectation of 50%) reflects the right-skewed distribution of realized volatility: rare but extreme high-vol episodes pull the trailing mean upward, leaving more days classified below the mean than above it. This is a property of vol's empirical distribution, not a methodological flaw.

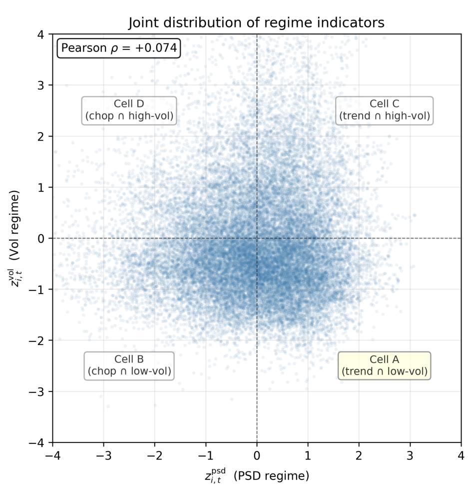

Figure 2: Joint distribution of PSD z-score and vol z-score across all (date, asset) cells. The cloud's circular symmetry around the origin is the visual signature of statistical independence between the two regime classifiers. Quadrants are labeled with the cells defined in Section 4.3.

There is no diagonal structure that would indicate the two indicators carry redundant information. This independence is the structural fact on which the rest of the analysis rests. It implies that conditioning on PSD regime and conditioning on vol regime are operationally distinct interventions, and that combining them partitions the (date, asset) state space into four cells of approximately equal occupancy.

4.2 Headline Strategy Comparison

Table 3 presents performance metrics for the four strategies over the full backtest period, January 2010 through the most recent available data (May 2026). All strategies are evaluated on net returns after transaction costs of 5 basis points per unit of turnover.

Strategy	Ann. Ret.	Ann. Vol.	Sharpe	Max DD	Calmar	Hit rate
Unconditional TSMOM	2.55%	6.06%	0.42	-13.5%	0.19	54.8%
PSD-filtered TSMOM	1.54%	3.97%	0.39	-10.5%	0.15	55.8%
Vol-filtered TSMOM	2.15%	4.25%	0.51	-8.0%	0.27	55.3%
Intersection-filtered	1.62%	2.71%	0.60	-4.6%	0.35	52.8%

Table 3: Performance metrics for the four strategies, full sample (January 2010 through 2026), net of 5 bps transaction costs. Absolute return levels are not directly comparable across strategies because they reflect different in-market exposures; risk-adjusted metrics (Sharpe, Calmar) are the meaningful basis for ranking.

Three findings are immediately visible. First, the vol filter improves on the unconditional baseline across every risk-adjusted metric: Sharpe rises from 0.42 to 0.51, maximum drawdown shrinks from -13.5% to −8.0%, and Calmar more than doubles from 0.19 to 0.27. This is the expected pattern from the volatility-scaled momentum literature [8]. Second, the PSD filter applied alone does not improve on the unconditional baseline. Sharpe falls to 0.39 and Calmar to 0.15. The PSD filter does reduce volatility and drawdown, but not enough to compensate for the reduction in return. Third, the intersection-filtered strategy produces the best risk-adjusted performance in the comparison set: Sharpe 0.60, Calmar 0.35, maximum drawdown of only -4.6%.

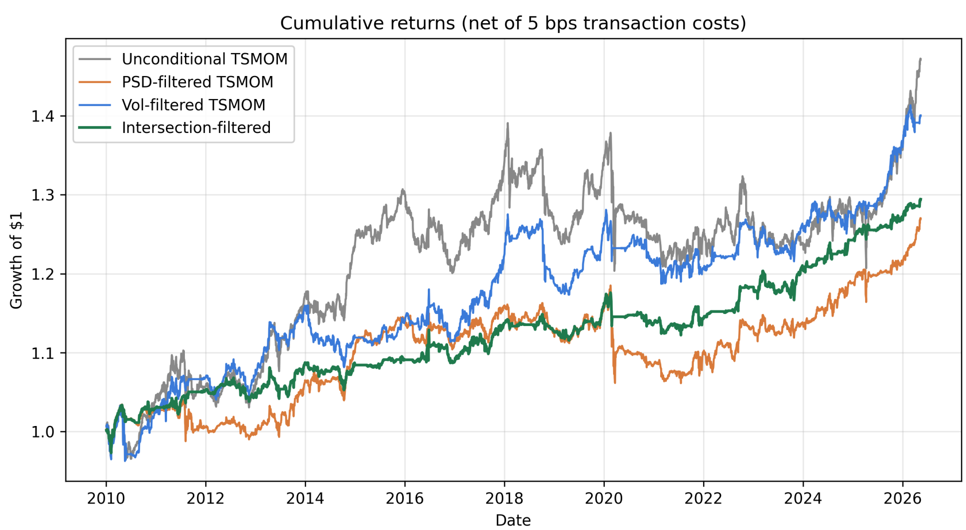

Figure 3

Figure 3 shows the cumulative return paths of the four strategies. The intersection strategy's path is visibly the smoothest, reflecting its lower volatility and shallower drawdowns. The unconditional strategy has the largest absolute drawdowns and the most variable path.

The fact that PSD filtering alone underperforms, yet combining it with the vol filter produces the best result in the set, is the empirical puzzle this section now resolves.

4.3 Cell-Conditional Decomposition

The independence of the two regime classifiers (Section 4.1) means that we can meaningfully partition the (date, asset) state space into four cells based on the joint regime classification. Following the quadrant labeling in Figure 2, the cells are:

A. Trending ∩ Low-vol: $z^{\text{psd}} > 0$ and $z^{\text{vol}} \leq 0$
B. Choppy ∩ Low-vol: $z^{\text{psd}} \leq 0$ and $z^{\text{vol}} \leq 0$
C. Trending ∩ High-vol: $z^{\text{psd}} > 0$ and $z^{\text{vol}} > 0$
D. Choppy ∩ High-vol: $z^{\text{psd}} \leq 0$ and $z^{\text{vol}} > 0$.

For each cell, I restricted the unconditional TSMOM strategy to only those (date, asset) pairs that fall in the cell. The asset is traded according to the momentum signal when in that cell, and held flat otherwise. The resulting Sharpe ratios isolate where in regime space momentum's risk-adjusted edge lives.

Metric	**Cell A** trend∩lowvol	**Cell B** chop∩lowvol	**Cell C** trend∩highvol	**Cell D** chop∩highvol
Occupancy	29.5%	28.0%	21.4%	21.1%
Annualized return	1.62%	0.39%	-0.16%	0.25%
Annualized vol	2.71%	2.62%	2.51%	2.08%
Sharpe	**0.60**	0.15	-0.06	0.12
Max drawdown	-4.6%	-9.7%	-12.1%	-9.8%
Calmar	0.35	0.04	-0.01	0.03

Table 4: Cell-conditional performance of the unconditional TSMOM strategy, restricted to days when the asset is in each of the four regime cells. Occupancy is the fraction of all (date, asset) pairs falling in the cell. Cell A, the intersection of trending and low-vol regimes, captures nearly all of momentum's risk-adjusted edge.

Table 4 delivers the central finding of the project. Cell A has Sharpe 0.60; the next-best cell (B) has Sharpe 0.15; cell C has *negative* Sharpe of -0.06; cell D has 0.12. The edge of momentum strategies on this universe is concentrated almost entirely in the trending ∩ low-volatility cell.

Figure 4 visualizes the same finding. Cell A's Sharpe is nearly four times that of the next-best cell, and the negative Sharpe of cell C is particularly informative.

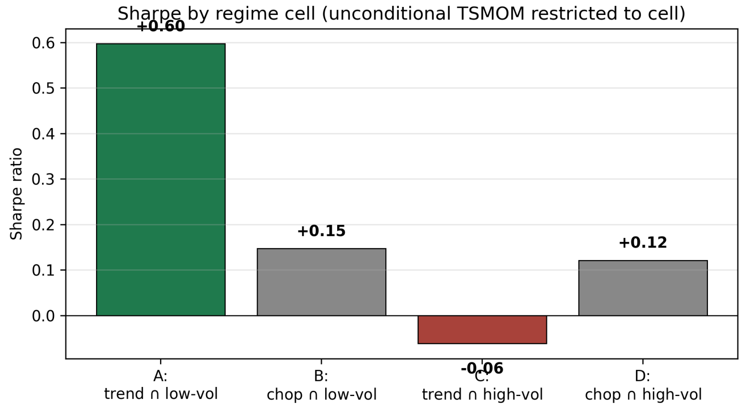

Figure 4: Sharpe ratios by regime cell, unconditional TSMOM restricted to each cell. Cell A (trending ∩ low-vol) captures essentially all of momentum's positive risk-adjusted edge. Cell C (trending ∩ high-vol) has negative Sharpe, which explains why the PSD filter applied alone, which admits both cells A and C, underperforms the baseline.

This decomposition solves the puzzle of why PSD filtering alone underperforms. The PSD filter admits cells A and C indiscriminately, since both are trending regimes. But Cell A has Sharpe 0.60 while cell C has Sharpe -0.06: the filter's gains in cell A are offset by losses in cell C. The vol filter admits cells A and B, both of which have positive Sharpe (0.60 and 0.15 respectively), so the vol filter improves on the baseline. The intersection filter admits only cell A, isolating the edge directly.

The mechanism is consistent with economic intuition. Short-lag positive autocorrelation in returns (the property the PSD ratio detects) reflects different underlying dynamics in low-vol and high-vol environments. In low-vol conditions, persistence typically reflects gentle, sustained directional drift, which the regime momentum strategies are designed to exploit. In high-vol conditions, persistence often reflects violent trending moves such as crashes or sharp recoveries, which tend to reverse at the horizons the 12-month minus 1-month momentum signal operates on. The PSD classifier, by design, does not distinguish between these two kinds of persistence; the vol filter does. Combining the two filters separates the persistence that helps momentum from the persistence that hurts it.

4.4 Statistical Tests of Sharpe Differences

Sharpe ratio differences between strategies are estimated with substantial sampling error. I applied the Memmel (2003) test (Equation 14) to the three most relevant pairwise comparisons.

Comparison	$\widehat{SR}_1$	$\widehat{SR}_2$	z	p-value
Intersection vs. Vol-filtered	0.60	0.50	0.60	0.55
Intersection vs. Unconditional	0.60	0.42	0.85	0.39
Vol-filtered vs. Unconditional	0.50	0.42	0.51	0.61

Table 5: Memmel (2003) test for Sharpe ratio differences between strategies, full sample. No pairwise difference reaches conventional significance thresholds. The standard error of the difference is inflated by the high correlation among strategies (all share the same underlying momentum signal), limiting the test's power.

None of the pairwise Sharpe differences reach conventional significance ($p < 0.05$). The intersection-versus-vol comparison, in particular, yields $z = 0.60$ with $p = 0.55$ i.e., the observed Sharpe gap of

0.10 is not distinguishable from zero given the available data and the high correlation between the two strategies' returns (both share the underlying momentum signal).

This is an important honest caveat to emphasize. The headline ranking (Intersection > Vol-filtered > Unconditional > PSD-filtered) is suggestive but not statistically definitive on the full sample. The strongest evidence for the intersection strategy comes not from this pairwise comparison but from the period-by-period stability presented in Section 5 and from the structural pattern in Table 4, neither of which depends on rejecting a null of equal Sharpes.

The wider methodological point is that Sharpe-difference tests have limited power on the sample sizes typical of finance backtests [1]. Even strategies with materially different true Sharpes are difficult to distinguish statistically without decades of data. I present these p-values for completeness and report point estimates, confidence intervals, and structural diagnostics in preference to relying on any single significance test.

5 Robustness

This section evaluates the headline findings against three robustness checks: period-by-period stability, the consistency of the cell-conditional decomposition across periods, and sensitivity to the pre-registered parameter values.

5.1 Period-by-Period Stability

I evaluated each strategy across seven period slices: the full sample, a chronological in-sample/out-of-sample split, three five-year windows, and the full sample with 2020–2022 excluded. Table 6 reports Sharpe across all period slices.

Period	Unconditional	PSD	Vol	Intersection
Full sample (2010–2026)	0.35	0.32	0.47	**0.56**
In-sample 2010–2017	0.68	0.50	0.58	**0.56**
OOS 2018–2026	0.06	0.16	0.36	**0.56**
2010–2014	0.74	0.49	0.51	**0.58**
2015–2019	0.32	0.30	0.57	**0.51**
2020–2024	-0.15	0.15	0.08	**0.53**
Excl. 2020–2022	0.60	0.54	0.60	**0.66**

Table 6

The intersection strategy delivers Sharpe between 0.51 and 0.66 across all seven slices; a range of only 0.15 while every other strategy shows substantially more period dependence. On the in-sample/OOS split it is the only strategy that does not materially degrade: 0.56 in-sample, 0.56 out-of-sample.

In 2020–2024, characterized by rapid regime transitions, the unconditional strategy delivered Sharpe -0.15 while the intersection delivered 0.53. The regime filter's protective effect was largest precisely where unconditional momentum failed worst. Excluding 2020–2022 entirely, the intersection Sharpe is 0.66, slightly higher than the full-sample value, which rules out the concern that the headline finding depends on that window. The mask-independence diagnostic (Section 4.1) is similarly stable across periods, with PSD/vol mask agreement ranging between 46.8% and 51.0%.

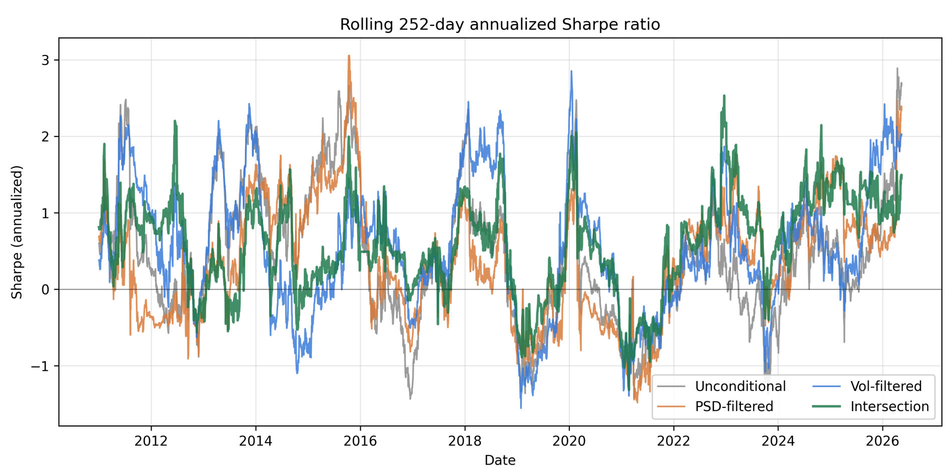

Figure 5

Figure 5 shows rolling 252-day Sharpe for each strategy and confirms the pattern visually: the intersection strategy exhibits the narrowest band and the fewest extended periods of negative rolling Sharpe.

5.2 Cell Structure Across Periods

The intersection strategy's stability traces directly to cell A's stability: across the six period slices, cell A's Sharpe varies only between 0.51 and 0.58. The other three cells are individually less stable. Cell D, for example, ranges from +0.92 in 2010–2014 to −0.30 in 2020–2024, and the relative ranking of cells B, C, D is not consistent across periods. The structural finding about cell A is therefore more robust than the secondary explanation for why PSD filtering alone underperforms; the latter holds on the full sample, the out-of-sample period, and four of the six sub-periods, but not cleanly on 2010–2014. The per-period cell Sharpe table is reported in Section B.

5.3 Parameter Sensitivity

Table 7 reports intersection-strategy Sharpe across the pre-registered grid of PSD window lengths and cutoff periods.

	Cutoff period (days)			
PSD window	15	20	25	30
40 days	0.34	0.49	0.49	0.49
60 days	0.56	**0.60**	0.60	0.65
90 days	0.35	0.46	0.50	0.50
120 days	0.34	0.44	0.49	0.34

Table 7: Intersection-strategy Sharpe across the parameter sensitivity grid. Pre-registered configuration (window = 60, cutoff = 20) shown in bold. Grid range: 0.34 to 0.65; median 0.49.

The grid exhibits a structural pattern consistent with spectral estimation theory. The 60-day window outperforms uniformly: all four cutoff values in the 60-day row exceed 0.55. Shorter windows (40 days) cannot resolve cycles longer than the window itself; longer windows (120 days) dilute regime transitions across multiple months. Cutoffs below 20 days underperform across all windows because they admit cycles shorter than the rebalance horizon into the "low-frequency" band, contaminating the classifier. The pre-registered configuration is near the top of the grid but is not the maximum: window 60 with cutoff 30 yields Sharpe 0.65.

The honest interpretation is mixed. The pattern in the grid is consistent with the pre-registration reasoning, supporting the specific choices made. However, only the 60-day window produces Sharpes that clearly exceed the vol-filtered baseline (0.51); outside this window length, the intersection does not dominate the simpler vol filter. The headline finding is therefore mechanistically explainable but sensitive to window length, and a wider grid between 50 and 70 days would help characterize the precise width of the favorable region. This sensitivity is a substantive limitation worth disclosing rather than dismissing.

6 Discussion

The empirical results admit a coherent interpretation. This section articulates the mechanism that links the independence of regime classifiers to the cell-conditional decomposition, and from there to the period stability of the intersection strategy.

6.1 Why PSD Filtering Alone Underperforms

The cell-conditional analysis (Section 4.3) shows that momentum's risk-adjusted edge is concentrated in cell A and that cell C, i.e., trending behavior in a high-volatility environment, has negative Sharpe of -0.06 on the full sample. The PSD filter does not distinguish between these two cells: both are classified as "trending" by the filter, because both share the property that returns exhibit short-lag positive autocorrelation. The filter therefore admits cells A and C in equal proportion, and the gains from concentrating on cell A are partially offset by the losses in cell C. On the full sample, this offset is just large enough to leave the PSD-filtered strategy slightly below the unconditional baseline.

The economic interpretation is that short-lag positive autocorrelation encodes different underlying dynamics depending on the volatility regime. In low-vol conditions, persistence typically reflects sustained directional drift, i.e., the regime that monthly-rebalanced momentum is designed to exploit. In high-vol conditions, persistence often reflects violent trending moves such as crashes, panic-driven rallies, and sharp recoveries, which tend to reverse at the horizons that the 12-month minus 1-month momentum signal operates on. The PSD ratio is invariant to these distinctions by construction; the realized volatility level discriminates them. Combining both signals separates the persistence that helps momentum from the persistence that hurts it.

This explanation is partial. The cell pattern is robust on the full sample, the out-of-sample period, and most sub-periods, but cell C is not uniformly the worst across all periods (Section 5.2).

6.2 Why the Intersection Strategy is Stable

The intersection strategy's most striking property is not its absolute Sharpe but its stability across periods. On the in-sample/OOS split, on five-year sub-periods, and on the excluded-2020–2022 robustness check, the intersection Sharpe stays within 0.51 to 0.66. No other strategy in the comparison set comes close to this stability.

The mechanism is direct: the intersection strategy is operationally equivalent to trading only when the asset is in cell A, and cell A itself is stable across periods. Cell A's Sharpe ranges from 0.51 to 0.58 across the six sub-periods examined. The intersection strategy inherits this stability. The instability of the other cells, and of the single-filter strategies that mix them, does not propagate into the intersection strategy because the intersection mask excludes the unstable cells by construction.

This stability is the strongest empirical evidence for the intersection filter, and it is more informative than any single-period Sharpe comparison. A strategy whose edge appears uniformly across distinct market environments is much less likely to be a statistical artifact than one whose edge is concentrated in a single favorable period. The Memmel test ($p > 0.39$ across pairwise comparisons) cannot reject equality of strategies on a single sample; the period stability provides the complementary evidence that the difference is structural rather than sampling noise.

6.3 What the Independence Finding Means

The empirical independence of PSD and volatility regimes (Section 4.1) is the structural fact that makes the rest of the analysis coherent. It implies that volatility-based regime detection, i.e., the standard approach, formalized in [8] via volatility-scaled momentum, does not exhaust the information available from each asset's own return history. The frequency-domain signal captures aspects of return dynamics that realized volatility does not, even though it does not, on its own, identify a profitable filter.

This finding is more nuanced than either a clean positive or a clean negative result on the original research question. The frequency-domain signal is not redundant with volatility, but it is also not, on its own, a useful filter for momentum. Its usefulness emerges only when combined with the volatility filter, in which case it identifies a subset of the state space where momentum's edge is concentrated.

The interpretation suggests a generalizable principle. Regime indicators that are individually weak filters can become useful in combination when they are statistically independent and when the underlying edge is concentrated in a corner of their joint distribution. The same logic would apply to other pairs of orthogonal regime classifiers such as macroeconomic versus market-internal, cross-sectional versus time-series, slow-moving versus fast-moving. Whether such combinations work depends on whether the edge to be filtered is similarly concentrated; this is an empirical question for each strategy and universe, not a general property of regime combinations.

7 Limitations

I acknowledged several limitations of the analysis. The first three are substantive constraints on the generalizability of the findings; the remainder are methodological caveats that bound what the results can be read to support.

Universe scope. The analysis is conducted on a 10-asset ETF universe spanning U.S. equities, international equities, Treasuries, commodities, and real estate, but excluding G10 currencies and using ETF rather than futures exposures. Time-series momentum is generally documented to deliver stronger Sharpe ratios on futures universes [6], in part because futures permit cheaper exposure to commodities and currencies and have longer continuous histories. I cannot rule out that the intersection-filter result would not generalize to a futures universe, where the dynamics of trending behavior may differ. A natural extension is to repeat the analysis on a liquid futures universe including FX.

Statistical precision. Sharpe-ratio estimates carry substantial sampling error on the sample sizes available to this study. Applying the standard approximation [1], the 95% confidence interval around the intersection strategy's full-sample Sharpe of 0.60 spans approximately ± 0.5 on roughly 16 years of data. The Memmel test for Sharpe differences (Table 5) fails to reject the null of equal Sharpes at conventional levels for any of the pairwise comparisons in the comparison set. The primary defense against this limitation is structural: the cell-conditional decomposition and the period stability are not single-Sharpe comparisons and do not depend on rejecting any one null. My specific point estimates should be read as central tendencies with wide associated uncertainty, not as precisely measured quantities.

Stability of the explanatory story. The interpretations in Section 6.1 that the PSD filter underperforms because it admits cell C, which has negative Sharpe, holds on the full sample, the out-of-sample period, and four of the six sub-periods examined, but not cleanly in 2010–2014. Cell A's stability across periods is robust; the relative ranking of cells B, C, and D is not. This can be interpreted as describing average behavior in most regimes, not a universal property of momentum strategies.

Parameter sensitivity. The headline finding is sensitive to the PSD window length: only the 60-day window produces intersection Sharpes that clearly exceed the vol-filtered baseline across cutoff values (Section 5.3). The grid pattern is consistent with the bias-variance argument used to justify the 60-day pre-registration, but do note that the pre-registered configuration sits near the top of the grid. A wider sensitivity analysis, including windows between 50 and 70 days, would help characterize the precise width of the favorable region.

Implementation assumptions. The backtest assumes transaction costs of 5 basis points per unit of turnover, no borrow costs on short positions, monthly rebalancing implemented at the closing price of the last trading day, and equal capital allocation across asset slots. I did not stress higher transaction costs (e.g., 10 or 15 bps), nor did I model short-borrow rebate rates, which can be material for sustained short positions in equity ETFs. The monthly rebalance choice also imposes a structural delay of up to 30 days between a regime classification change and the corresponding position adjustment; a daily-rebalance variant would respond faster but at substantially higher turnover.

8 Conclusion and Future Work

Through this project, I asked whether a frequency-domain regime indicator, derived from the rolling power spectral density of returns, adds information beyond a volatility-based regime indicator for the purpose of filtering a time-series momentum strategy. On a 10-asset ETF universe spanning 2010 to 2026, the answer is structured and not very simple. The two regime classifiers are empirically independent as they agree on approximately 50% of (date, asset) cells and exhibit per-asset correlations close to zero, yet neither filter, applied alone, delivers risk-adjusted performance superior to the simpler vol-filtered strategy. But, the intersection of the two filters does. The intersection-filtered strategy delivers Sharpe between 0.51 and 0.66 across in-sample, out-of-sample, and sub-period evaluations, while every other strategy in the comparison set exhibits substantially more period dependence.

The mechanism underlying this finding is a cell-conditional concentration of momentum's risk-adjusted edge. When the (date, asset) state space is partitioned by the two binary regime classifiers, momentum's edge is concentrated in the cell where the asset is simultaneously classified as trending by the spectral filter and as low-volatility by the realized volatility filter. The intersection strategy isolates this cell directly.

Several natural extensions follow: The most direct is to repeat the analysis on a liquid futures universe, where time-series momentum has historically delivered stronger Sharpe ratios. A second extension is to test the same intersection construction with mean-reversion as the base strategy rather than momentum, on the conjecture that the favorable cell for mean-reversion may be the diagonally opposite corner (choppy $\cap$ high-vol) of the regime partition. A third is to formalize the cell-conditional analysis with a permutation test on the regime labels, which would provide statistical evidence for the cell A dominance independent of any single-period Sharpe comparison. A fourth is to widen the parameter sensitivity grid to characterize the precise window-length range over which the intersection result holds.

The most general implication is that, regime indicators that are individually weak filters can become useful in combination when they are statistically independent and when the underlying edge is concentrated in a corner of their joint distribution. The independence diagnostic introduced in Section 4.1 is a simple empirical tool for evaluating whether such a combination is likely to add value, and may be useful in evaluating other proposed regime indicators in future work.

8.1 Code Availability

All code implementing the pipeline described in this section is available at
`https://github.com/lilNewbie/regime-filtered-tsmom`.

References

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A Pre-Registration Document

The following document was finalized prior to any backtesting being performed. It is reproduced here verbatim to support the pre-registration claim in Section 3.4.

Research Question

Does a frequency-domain regime indicator (low/high PSD ratio on returns) improve a time-series momentum strategy, relative to (a) unconditional TSMOM and (b) TSMOM filtered by realized volatility?

Hypotheses

- H1: PSD-filtered TSMOM has higher Sharpe than unconditional TSMOM.
- H2: PSD-filtered TSMOM has higher Sharpe than vol-filtered TSMOM.
- Null (either): no significant difference.

Universe

SPY, QQQ, IWM, EFA, EEM, TLT, IEF, GLD, DBC, VNQ; equal-weighted across all asset slots after individual position sizing.

Parameters

- Data frequency: daily adjusted closes, 2007-01-01 to present
- Momentum signal: sign of trailing 12-month minus trailing 1-month log return
- Rebalance frequency: monthly (last trading day)
- Position sizing: vol-scaled to 10% annualized target per asset
- Volatility estimator: 60-day rolling standard deviation of daily returns, annualized
- PSD method: Welch's method, nperseg = 60, noverlap = 30, linear detrending
- PSD window: 60 trading days of daily returns
- Frequency cutoff: period = 20 trading days; low band: periods $>$ 20 days; high band: periods $\leq$ 20 days
- PSD feature: $\log\left(\sum L \,/\, \sum H\right)$
- Regime classifier: rolling z-score of the feature over past 252 days; trending if $z > 0$
- Vol-filter baseline: rolling z-score of 20-day realized vol over past 252 days; low-vol if $z \leq 0$

Backtest Protocol

- Walk-forward start: 2010-01-01 (first 3 years used for indicator warm-up)
- No parameter re-fitting inside the backtest; all parameters fixed above
- Transaction cost assumption: 5 bps per unit of turnover
- Out-of-sample metrics: annualized return, annualized vol, Sharpe, max drawdown, Calmar, hit rate (% of positive months), turnover

Decision Rules

- If H1 and H2 both fail: report as primary finding ("PSD filter does not add value")
- If H1 passes but H2 fails: report as "filter helps but no better than vol filter"
- If both pass: report effect size and robustness to parameter perturbations

Sensitivity Checks (appendix)

- PSD window: $\{40, 60, 90, 120\}$
- Frequency cutoff period: $\{15, 20, 25, 30\}$
- Sub-period breakdown: 2010–2014, 2015–2019, 2020–2024
- Exclude 2020–2022 (the trending bear market period): does headline result survive?

B Per-Period Cell Sharpes

Cell Sharpes from Section 5.2, reported across all period slices for reference.

Period	**Cell A** trend∩lowvol	**Cell B** chop∩lowvol	**Cell C** trend∩highvol	**Cell D** chop∩highvol
Full sample (2010–2026)	**0.56**	0.13	-0.13	0.05
In-sample 2010–2017	**0.56**	0.29	0.09	0.46
OOS 2018–2026	**0.56**	-0.04	-0.28	-0.21
2010–2014	0.58	0.18	0.10	**0.92**
2015–2019	**0.51**	0.37	-0.22	-0.29
2020–2024	**0.53**	-0.53	-0.26	-0.30

Table 8: Cell Sharpe ratios across period slices. Cell A's Sharpe varies only between 0.51 and 0.58 across all periods. The relative ranking of cells B, C, and D is not stable: cell D dominates in 2010–2014 but underperforms in later periods.

C Statistical Tests of Sharpe Differences

I reported the Memmel (2003) test for pairwise Sharpe-ratio differences referenced in Section 4.4.

Comparison	$\widehat{SR}_1$	$\widehat{SR}_2$	z-statistic	p-value
Intersection vs. Vol-filtered	0.60	0.50	0.60	0.55
Intersection vs. Unconditional	0.60	0.42	0.85	0.39
Vol-filtered vs. Unconditional	0.50	0.42	0.51	0.61

Table 9: Memmel (2003) test for pairwise Sharpe-ratio differences, full-sample. Sample size $n = 4{,}115$ daily observations. No pairwise comparison reaches conventional significance levels.

The test's failure to reject equality reflects two factors: limited sample size by the standards of Sharpe estimation (approximately 16 years), and the high correlation among strategies, which all share the same underlying TSMOM signal and differ only in the regime mask. The correlation between strategy returns inflates the standard error of the Sharpe difference, reducing test power.

The primary evidence for strategy ranking comes from the period-stability and cell-conditional analyses (Section 5) rather than from any single pairwise test.

D Parameter Sensitivity Grid (Full Data)

The full parameter sensitivity grid from Section 5.3, including the PSD-only Sharpe and intersection maximum drawdown across all 16 configurations.

	Cutoff period (days)			
PSD window	15	20	25	30
Panel A: Intersection-strategy Sharpe				
40 days	0.34	0.49	0.49	0.49
60 days	0.56	**0.60**	0.60	0.65
90 days	0.35	0.46	0.50	0.50
120 days	0.34	0.44	0.49	0.34
Panel B: PSD-only-strategy Sharpe				
40 days	0.17	0.32	0.32	0.32
60 days	0.29	**0.39**	0.39	0.43
90 days	0.20	0.29	0.36	0.31
120 days	0.23	0.31	0.36	0.20
Panel C: Intersection-strategy max drawdown				
40 days	-6.7%	-5.4%	-5.4%	-5.4%
60 days	-4.8%	**-4.6**%	-4.6%	-4.2%
90 days	-5.5%	-5.1%	-5.3%	-4.8%
120 days	-5.2%	-6.2%	-6.4%	-5.8%

Table 10: Full parameter sensitivity grid. Pre-registered configuration (window 60, cutoff 20) shown in bold in each panel. Panel A reports intersection-strategy Sharpe; panel B reports PSD-only-strategy Sharpe for comparison; panel C reports maximum drawdown of the intersection strategy.